

FAYÇAL GHOZZI

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PROFILE

Fullstack software engineer with over 3 years of experience designing backend APIs, web/mobile interfaces, and critical banking integrations. Equally comfortable with server-side architecture (**Python, FastAPI, PHP, Node.js**) and interface development (**React, Next.js, React Native, Vue.js**).

I designed a payment middleware end-to-end for a Tunisian bank, took over and stabilized critical banking applications left in a fragile state, and I am currently developing the backend of a multimodal mobility platform processing over 20 million lines of data per day.

Author of the open-source framework LazyKitty.

PROFESSIONAL EXPERIENCE

R&D Software Engineer, CESI Strasbourg | Mon Trajet Vert JUL 2025 - present, Strasbourg, France

- Lead engineer on “Mon Trajet Vert”, aggregating GTFS/GTFS-RT feeds from 7 cities (Strasbourg, Paris, Nantes, Lyon, Toulouse, Bordeaux, La Rochelle)
- Designed a FastAPI backend on GCP Cloud Run implementing the RAPTOR algorithm for multimodal route calculation (tram, metro, bus, train, dockless bike-share via GBFS, personal bike with bike-parking search, wheelchair-accessible routes), complemented by an OSM pedestrian graph.
- Optimized the GTFS ingestion pipeline (20M+ lines): reduced initialization time from 49 hours to under 10 minutes.
- Reduced RAPTOR latency from 1 minute to under 6 seconds for large cities (a few milliseconds for medium/small cities)
- Developed a Dijkstra-based benchmarking tool to validate RAPTOR's performance.
- Integrated Firestore via a GraphQL API; handled GTFS edge cases (overnight trips, public holidays, line-color normalization)
- Coordinated weekly sprints with *Galadrim* (frontend partner), mentored interns and PhD students.

Stack: Python, FastAPI, React, Docker, GCP Cloud Run, Firestore, GraphQL, GTFS/GTFS-RT, C#

Software Engineer, Banque Tuniso-Libyenne (BTL) OCT 2024 - JUN 2025, Tunis, Tunisia

- End-to-end design of the BTL payment middleware Python/FastAPI: REST API consumed by the national centralized check-clearing platform (PECC), SOAP/OFS messages to Temenos T24 R11, OTP management, and two-level logging (global daily monitoring + operation-by-operation traceability)
- Took over and overhauled two critical banking applications (SWIFT operations management: MT103, MT202, MT202COV, MT950; digitalization of approval workflows for clearing, check reservation, and loans): introduced reusable services, exception handling, and clear error messages.
- Technical liaison between BTL management and Pixel Trade (external vendor) for BTL M Banking (Android/iOS) and the website: bug qualification and tracking
- Provided logs and reports to risk and audit teams: tracking of SWIFT operations, traceability of validations and accounting entries.

Stack: Python, FastAPI, PHP, COBOL, React, MySQL, Temenos T24, SOAP, REST API, JBase, TailwindCSS

Software Engineer, **Hortensia Agency**

MAY 2024 - OCT 2024, Remote

- Developed and stabilized features on Project Curve, a financial modeling platform based on sigmoid functions, using React.js and Flask
- Integrated Stripe payments: managed free/paid subscriptions and the billing cycle via webhooks

Stack: Python, Flask, JAVA, Spring Boot, C#, .NET, React, Vue.js Stripe, MongoDB

Software Engineer, **7Systems GmbH**

SEP 2022 - APR 2024, Germany (Hybrid)

- Contributed to the development and stabilization of the bmacloud.de SaaS platform.
- Migrated the legacy frontend to React.js; improved team productivity by 30% through test automation and CI/CD improvements.

Stack: React, PHP 7.4, MySQL

PERSONAL PROJECTS

- **LazyKitty** - Open-source reactive Python web framework: full-stack interfaces via persistent WebSockets, no JavaScript or HTML templates, Next.js-style routing (<https://www.lazykitty.dev>)
- **Real-time drone perception system:** custom YOLOv11 + TensorRT on NVIDIA Jetson Orin, ROS2 pipeline, React/TypeScript operator dashboard.
- **Real-time GPS tracking system:** Android application (React Native), Leaflet web dashboard, JWT authentication, Docker deployment.
- **Tic-Tac-Two:** real-time online multiplayer game (Next.js, WebSockets) - (<https://tic-tac-two-phi.vercel.app>)

KEY SKILLS

- **Backend:** Python, FastAPI, Flask, PHP, Laravel, Node.js, Java, REST, SOAP, GraphQL, C#, .NET
- **Frontend & Mobile:** React, Next.js, React Native, Vue.js, TypeScript, JavaScript, Kotlin, Symfony
- **Data, Infra & Security:** Docker, GCP, PostgreSQL, MySQL, MongoDB, Firestore, JWT, GTFS/GTFS-RT, ROS2, TensorRT
- **Banking / Fintech:** Temenos T24, messages SWIFT (MT103, MT202, MT202COV, MT950, among others), COBOL, Stripe
- **Languages:** English (Fluent), French (Fluent), German (basic)

EDUCATION

École Supérieure Privée d'Ingénierie et de Technologies (ESPRIT) – National Engineering Degree in Computer Science, 2017-2022

CERTIFICATIONS

- Artificial Intelligence Course, Samsung Innovation Campus - February 2025
- Fundamentals of Deep Learning, NVIDIA - October 2021
- Intermediate Python, DataCamp - January 2022

PORTFOLIO

